

DATABASE PROJECT PORTFOLIO

Retail Shop Database System — Pasar Mini Mohd. Salleh

ADAM ISKANDAR BIN NAZRI

Student ID: 2024277906

Faculty of Computer and Mathematical Sciences

Universiti Teknologi MARA (UiTM)

Contact	01111263076 1603adam@gmail.com
Course	ICT200 – Introduction to Database Management
Year	2025 – 2026

1. About Me

I am Adam Iskandar bin Nazri, a Computer Science student at UiTM Jasin. I am passionate about database design, data management, and building structured systems that solve real-world business problems. This portfolio highlights my hands-on experience designing and implementing a relational database system for a real campus mini-market.

Core Competencies - Relational Database Design - SQL (DDL & DML) - ER Diagram & Normalization (3NF) - MySQL Workbench - Data Analysis & Reporting	Personal Details Role in Project: Project Manager Hometown: Melaka Tengah, Melaka Email: 1603adam@gmail.com Phone: 01111263076
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2. Project Overview

Company Background

Pasar Mini Mohd. Salleh (also known as Gerai 24) is a campus convenience store operating within UiTM Jasin. The store functions similarly to a 7-Eleven, serving the daily needs of students and faculty with ready-to-eat items, beverages, snacks, and essential academic supplies.



Gerai 24 – Pasar Mini Mohd. Salleh, UiTM Jasin Campus

Problem Statement

Before the database system was implemented, the store relied entirely on manual paper-based records. This resulted in three major data management issues:

- Lengthy development time, manual files slowed record creation, filing, and retrieval
- Data entry errors, human mistakes in stock counts led to expired goods and restocking failures
- Lack of security, no constraints on who could modify records, enabling fraud and data tampering

My Role

As Project Manager, I led the planning and coordination of the entire database project, including requirements gathering (direct interview with Encik Zulfadli, the owner), task delegation, timeline management, and final review of all database components.

3. Database Design

Entity Relationship Diagram (ERD)

The database consists of 9 normalized tables designed using Crow's Foot notation, covering all business operations of the store.

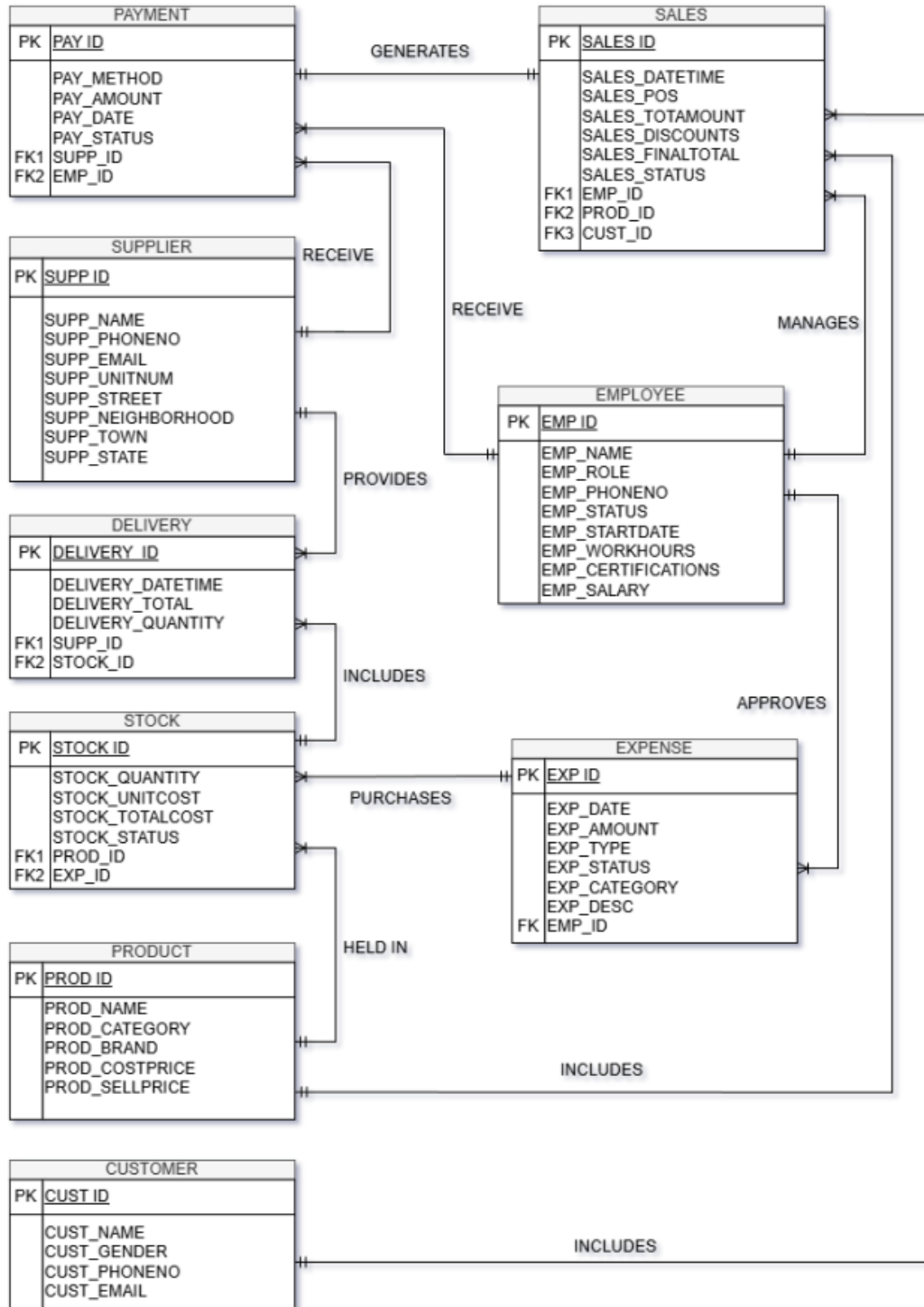


Figure 1: Finalized ERD – Crow's Foot Notation

Tables & Relationships

Table	Purpose	Records
CUSTOMER	Stores customer loyalty and transaction details	20 records
EMPLOYEE	Personnel records, payroll, and scheduling	6 records
PRODUCT	Master product catalogue with pricing	139 records
SALES	Point-of-sale transaction log	25 records
PAYMENT	Payment method and transaction tracking	25 records
SUPPLIER	Vendor contact and address details	104 records
DELIVERY	Goods received from suppliers	25 records
STOCK	Inventory quantity and cost tracking	25 records
EXPENSE	Non-inventory business expenditures	25 records

4. SQL Implementation

Data Definition Language (DDL)

All tables were created using properly structured DDL statements with primary keys, foreign keys, NOT NULL constraints, and appropriate data types to enforce data integrity.

9.2 Create table commands for all tables

```
CREATE TABLE employee (  
    emp_id INT AUTO_INCREMENT PRIMARY KEY,  
    emp_name VARCHAR(100) NOT NULL,  
    emp_role VARCHAR(50),  
    emp_phoneno VARCHAR(20),  
    emp_status VARCHAR(20),  
    emp_startdate DATE,  
    emp_workhours DECIMAL(5,2),  
    emp_certifications TEXT,  
    emp_salary DECIMAL(10,2)  
);  
  
CREATE TABLE customer (  
    cust_id INT AUTO_INCREMENT PRIMARY KEY,  
    cust_name VARCHAR(100) NOT NULL,  
    cust_gender CHAR(1),  
    cust_phoneno VARCHAR(20),  
    cust_email VARCHAR(100)  
);  
  
CREATE TABLE product (  
    prod_id INT AUTO_INCREMENT PRIMARY KEY,  
    prod_name VARCHAR(100) NOT NULL,  
    prod_category VARCHAR(50),  
    prod_brand VARCHAR(50),  
    prod_costprice DECIMAL(10,2),  
    prod_sellprice DECIMAL(10,2)  
);
```

Figure 2: DDL Statements for Table Creation in MySQL Workbench

Data Manipulation Language (DML) — Sample Queries

The following queries demonstrate the range of SQL skills applied in this project:

Query 1: Retrieving Customer Records

10.1 Display all records from all tables

Question

Displaying table customer

Screenshot of SQL Command

```
SELECT * FROM customer;
```

Screenshot of Output

cust_id	cust_name	cust_gender	cust_phoneno	cust_email
101	Ahmad bin Hassan	M	0123456789	ahmad.hassan@gmail.com
102	Siti Nurhaliza	F	0198765432	siti.nur@yahoo.com
103	Muhammad Hafiz	M	0167894561	hafiz.m@gmail.com
104	Nur Aisyah	F	0134567890	aisyah.nur@outlook.com
105	Azman bin Osman	M	0145678901	azman.osman@hotmail.com
106	Nurul Ain	F	0156789012	nurul.ain@gmail.com
107	Khairul Anwar	M	0187890123	khairul.a@yahoo.com
108	Fatimah Zahra	F	0129012345	fatimah.z@outlook.com
109	Mohd Faiz	M	0178901234	faiz.mohd@gmail.com
110	Zainab binti Ali	F	0169012345	zainab.ali@hotmail.com
111	Ismail bin Ahmad	M	0190123456	ismail.ahmad@yahoo.com
112	Aishah Maisarah	F	0121234567	aishah.m@gmail.com
113	Haris Fadhilah	M	0132345678	haris.f@outlook.com
114	Sarah binti Abdullah	F	0143456789	sarah.abdullah@gmail.com
115	Razak bin Hamid	M	0154567890	razak.hamid@yahoo.com
116	Amirul Hakim	M	0172345678	amirul.hakim@uitm.edu.my
117	Nadira Syafinaz	F	0128765432	nadira.syafinaz@uitm.edu....
118	Fikri Azlan	M	0193456789	fikri.azlan@gmail.com
119	Hani Binti Rosli	F	0135678901	hani.rosli@uitm.edu.my
120	Aizat Bin Zainal	M	0146789012	aizat.zainal@hotmail.com

Screenshot of the Number of Records

09:35:57 SELECT * FROM customer LIMIT 0, 1000

20 row(s) returned

Figure 3: SELECT query output displaying 20 customer records

Query 2: Column Functions – Employee Expense Summary

10.5 List significant information that uses the column functions clause

Question

Determine the total amount of expenses approved by each employee. Display the employee's name, role, and phone number.

Screenshot of SQL Command

```
SELECT emp_name, emp_role, emp_phoneno, SUM(exp_amount) as TotalExpense
FROM employee
JOIN expense ON employee.emp_id = expense.emp_id
GROUP BY emp_name, emp_role, emp_phoneno;
```

Screenshot of Output

emp_name	emp_role	emp_phoneno	TotalExpense
Muhammad Zulfadli	Manager	0128457391	7700.00
Nurul Aini	Assistant Manager 1	0137682945	1695.50
Hassan	Assistant Manager 2	0145938267	2400.00
Hussin	Supervisor	0116748392	3055.20
Farah Adilah	Accountant	0163829457	255.00
Nurul Syafie	Part-Time Staff	0174928365	420.50

Screenshot of the Number of Records

```
21:48:56 SELECT emp_name, emp_role, emp_phoneno, SUM(exp_amount) as TotalExpense FROM employee JOIN expense ON employee.emp_id = expense.e... 6 row(s) returned
```

Figure 4: SUM aggregate function grouping total expenses by employee

Query 3: JOIN Query – Customers and Sales

10.8 List significant information that uses the join clause with other tables

Question

Display the names and phone numbers of customers along with their total purchase amounts.

Screenshot of SQL Command

```
SELECT cust_name, cust_phoneno, sales_finaltotal, sales_discounts
FROM customer
JOIN sales ON customer.cust_id = sales.cust_id;
```

Screenshot of Output

cust_name	cust_phoneno	sales_finaltotal	sales_discounts
Siti Nurhaliza	0198765432	45.00	0.00
Zainab binti Ali	0169012345	115.00	5.50
Hani Binti Rosli	0135678901	15.00	0.00
Aizat Bin Zainal	0146789012	75.00	10.00
Ahmad bin Hassan	0123456789	32.00	0.00
Ahmad bin Hassan	0123456789	53.00	2.20
Razak bin Hamid	0154567890	10.00	0.00
Sarah binti Abdullah	0143456789	180.00	20.00
Nurul Ain	0156789012	60.00	5.00
Mohd Faiz	0178901234	42.00	0.00
Amirul Hakim	0172345678	25.00	0.00
Ismail bin Ahmad	0190123456	90.00	8.00
Ahmad bin Hassan	0123456789	27.00	3.00
Nur Aisyah	0134567890	50.00	0.00
Razak bin Hamid	0154567890	15.50	0.00
Amirul Hakim	0172345678	100.00	10.00
Haris Fadhillah	0132345678	45.00	0.00
Ahmad bin Hassan	0123456789	22.00	0.00
Nurul Ain	0156789012	70.00	5.00
Muhammad Hafiz	0167894561	12.00	0.00
Aizat Bin Zainal	0146789012	125.00	15.00
Fikri Azlan	0193456789	38.00	0.00
Aishah Maisarah	0121234567	55.00	5.00
Sarah binti Abdullah	0143456789	28.00	0.00
Nurul Ain	0156789012	225.00	25.00

Screenshot of the Number of Records

```
22:18:04 SELECT cust_name, cust_phoneno, sales_finaltotal, sales_discounts FROM customer JOIN sales ON customer.cust_id = sales.cust_id LIMIT 0, 1000 25 row(s) returned
```

Figure 5: INNER JOIN linking customer table with sales data

5. Technical Skills Demonstrated

Skill Area	What I Did
Database Design	Designed 9-table normalized ERD using Crow's Foot notation up to 3NF
DDL – CREATE/DROP	Wrote all CREATE TABLE statements with constraints and foreign keys
DML – SELECT	Wrote 15+ queries including single-table, multi-table, and aggregate queries
JOIN Queries	Used INNER JOIN, LEFT JOIN across multiple tables (sales, customer, product)
Aggregate Functions	Used SUM(), AVG(), COUNT(), MAX() with GROUP BY and HAVING clauses
Subqueries	Used NOT IN subqueries to identify uncontacted suppliers
Filtering	Applied WHERE, LIKE, IN, comparison operators for data filtering
MySQL Workbench	Used as primary tool for database creation, execution, and output capture
Project Management	Coordinated team of 4, interviewed business owner, managed deadlines

Thank you for your time and consideration.

I look forward to the opportunity to contribute and grow with your organization.

Adam Iskandar bin Nazri

01111263076 | 1603adam@gmail.com